Alessandro Baccarini

Contact Information

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	Scholar	1334 citations, Aug. 2026

Research Interests

My research interests span applied cryptography, information security, and privacy, with an emphasis on distributed cryptographic protocols, privacy-preserving computation, and blockchain-based systems. A central theme of my work is bridging theory and practice: I design secure and efficient systems and protocols that connect rigorous cryptographic foundations with real-world applications, particularly in decentralized and heterogeneous settings.

Education

PhD, Computer Science, University at Buffalo — Aug. 2024
Advisor: Marina Blanton

MS, Cybersecurity, Fordham University — May 2019
Advisor: Thaier Hayajneh

BS, Physics, Fordham University — May 2017
Minor in Mathematics

Work Experience

Chief Cryptographer, Pogun — June 2026 – Present
Input Output Group

Applied Cryptographer, Partner Chains, Pogun — Sep. 2025 – June 2026
Input Output Group

Principal Consultant, Cryptography — Sep. 2024 – Present
Guardian Cryptography

Research Assistant, Computer Science — Jun. 2019 – Aug. 2024
University at Buffalo

Teaching Assistant, Computer Science — Jan. 2020 – May 2022
University at Buffalo

Adjunct Assistant Professor, Physics — Aug. 2017 – May 2019
Fordham University

Graduate Research Assistant, Cybersecurity — Aug. 2017 – May 2019
Fordham University

Awards and Recognition

Distinguished Artifact Reviewer, Privacy Enhancing Technologies Symposium (PETS) July 2026

Noteworthy Artifact Reviewer, USNEIX Security Symposium Aug. 2026

Alan Selman Scholarship, University at Buffalo Mar. 2024
First place $2000 cash prize, focus in theoretical computer science.

GSAS Centennial Scholarship, Fordham University Aug. 2017 – May 2019
Full tuition support and stipend (academic year + summer).

Experience

Pogun Bridge (Mirror) Dec. 2025 – Present
Input Output Group, *Pogun*

- Leading the cryptographic research, design, and execution of the *Pogun Mirror*, a trust-minimized bridge architecture between Bitcoin and Cardano based on SNARK witness encryption and secure multi-party computation (MPC).

Capacity Exchange Threshold Signature Sep. 2025 – Dec. 2025
Input Output Group, *Partner Chains*

- Led the cryptographic design of the Capacity Exchange for the Midnight blockchain, a decentralized platform that enables externally funded private transactions (from Bitcoin, Ethereum, and Cardano) on Midnight by leveraging threshold signature schemes (TSS) and zero-knowledge proofs.
- Evaluated state-of-the-art TSS constructions (FROST, MuSig2, and their post-quantum variants) based on their security guarantees, performance characteristics, and suitability for the Capacity Exchange ecosystem.
- Collaborated closely with internal and external R&D teams to identify opportunities for novel cryptographic and economic research and to design bespoke engineering and architectural solutions.

Principal Consultant Sep. 2024 – Present
Guardian Cryptography

- Provide expert consulting in applied cryptography, privacy-enhancing technologies, and secure software development for startups, research teams, and enterprise clients.

Threshold Decryption for FHE Sep. 2024 – Dec. 2024
Guardian Cryptography. *Client: Blockchain R&D Organization*

- Analyzed distributed threshold decryption protocols for multi-party fully homomorphic encryption (FHE) schemes with application to private smart contract deployment on Ethereum-like blockchains.
- Implemented and evaluated an actively secure threshold decryption construction based on Shamir secret sharing over Galois rings in C++, obtaining an up to 4× performance improvement over prior works.
- Designed a general-purpose threshold distributed key generation protocol compatible with various multi-party FHE schemes, and implemented it within the MP-SPDZ framework.

PICCO Compiler Jan. 2020 – Aug. 2024
University at Buffalo Repository

- Core developer and maintainer of PICCO, a source-to-source MPC compiler library for translating general-purpose programs into their secure distributed equivalents.
- Integrated ring-based constructions into the compiler to support general-purpose secure computation over diverse input domains and broaden application flexibility.

- Performed extensive optimizations to existing field-based protocols and led a large-scale refactor of over 100k lines of code to enhance long-term maintainability and enable support for stronger security models.
- Mentored undergraduate students on projects including optimizing and parallelizing networking layers across parties, and developing a web interface to facilitate secure input/output interactions.

MPC Framework for Privacy-Preserving AI Jan. 2020 – July 2023
University at Buffalo Repository

- Designed a novel comprehensive ring-based framework of replicated secret sharing MPC protocols for an arbitrary number of parties in the semi-honest, honest majority setting.
- Implemented protocols in C++, applying extensive profiling and low-level optimizations that led to up to 33× performance improvements over state-of-the-art secret sharing techniques.
- Applied MPC to privacy-preserving machine learning tasks, including (quantized) convolutional deep neural network inference and support vector machine classification.
- Discovered an algebraic optimization for secure quantized inference that minimizes the overall ring modulus across multiple layer evaluations, yielding over 2× performance improvement on average.

Secure Computation Information Disclosure Analysis Jan. 2021 – Present
University at Buffalo Repository

- Developed an information-theoretic technique to quantify leakage about private inputs from arbitrary secure computation outputs, enabling practical assessment of residual disclosure under complex function evaluation.
- Analyzed common statistical functions under practical MPC configurations and proposed concrete mitigation strategies for real-world deployment.
- Combined the methodology with entropy estimation techniques using machine learning to assess leakage from complex descriptive statistical measures (e.g., variance, order statistics).
- Awarded first place Alan Selman Scholarship in Theoretical Computer Science for this work, recognized for its combination of rigorous theory and real-world relevance to secure data analysis.

Publications

Thesis

[1] **Alessandro Baccarini**. *New Directions in Secure Multi-Party Computation: Techniques and Information Disclosure Analysis*. PhD thesis, University at Buffalo, 2024.

Conference Proceedings

[2] **Alessandro Baccarini**, Marina Blanton, and Shaofeng Zou. Understanding information disclosure from secure computation output: A study of average salary computation. In *ACM Conference on Data and Application Security and Privacy (CODASPY)*, pages 187–198, 2024.

[3] **Alessandro Baccarini** and Thaier Hayajneh. Evolution of format preserving encryption on IoT devices: FF1+. In *Hawaii International Conference on System Sciences (HICSS)*, pages 1628–1637, 2019.

[4] Abdullah Alhayajneh, **Alessandro Baccarini**, and Thaier Hayajneh. Quality of service analysis of VoIP services. In *IEEE Annual Ubiquitous Computing, Electronics & Mobile Communication Conference (UEMCON)*, pages 812–818, 2018.

Refereed Journals

[5] **Alessandro Baccarini**, Marina Blanton, and Shaofeng Zou. Understanding information disclosure from secure computation output: A comprehensive study of average salary computation. *ACM Transactions on Privacy and Security (TOPS)*, 28(1):1–36, 2024.

[6] **Alessandro Baccarini**, Marina Blanton, and Chen Yuan. Multi-party replicated secret sharing over a ring with applications to privacy-preserving machine learning. *Proceedings on Privacy Enhancing Technologies (PoPETs)*, 2023(1):608–626, 2023.

[7] Abdullah Alhayajneh, **Alessandro Baccarini**, Gary Weiss, Thaier Hayajneh, and Aydin Farajidavar. Biometric authentication and verification for medical cyber physical systems. *Electronics*, 7(12):436, 2018.

[8] Kristen Griggs, Olya Ossipova, Christopher Kohlios, **Alessandro Baccarini**, Emily Howson, and Thaier Hayajneh. Healthcare blockchain system using smart contracts for secure automated remote patient monitoring. *Journal of Medical Systems*, 42(7):130, 2018.

In Preparation

[9] **Alessandro Baccarini**, Marina Blanton, and Adithya Vadapalli. Private data streaming statistics, 2026.

[10] **Alessandro Baccarini**, Marina Blanton, and Shaofeng Zou. Simulation-based techniques for estimating information disclosure from secure function evaluations, 2026.

Technical Presentations

- Molekule Consulting, AI Security & Privacy. Virtual. — Sep. 2025
- RAND Corporation, Engineering & Applied Sciences Dept. Virtual. — Apr. 2025
- Intel Labs, Security & Privacy Research Group. Virtual. — Mar. 2025
- Riverside Research, Secure & Resilient Systems Group. Lexington, MA. — Feb. 2025
- MITRE Corporation, Cyber for Identity Trust & Assurance Dept. Virtual. — Jan. 2025
- ACM CODASPY 2024. Porto, Portugal. — June 2024
- PETS 2023. Lausanne, Switzerland. — July 2023
- Great Lakes Security Day, RIT. Rochester, NY. — Apr. 2023
- IEEE UEMCON 2018. New York, NY. — Nov. 2018

Professional Service

Conference Committees

Privacy Enhancing Technologies Symposium (PETS), artifact evaluation committee	2026
USENIX Security Symposium, artifact evaluation committee	2023, 2024, 2026
IEEE Symposium on Security and Privacy, poster jury	2025

Refereeing

Journal of Computer and System Sciences (JCSS)

IEEE Transactions on Information Forensics and Security (TIFS)

IEEE Transactions on Dependable and Secure Computing (TDSC)

European Symposium on Research in Computer Security (ESORICS)

IEEE/ACM International Conference on Automated Software Engineering (ASE)

Multidisciplinary Digital Publishing Institute (MDPI) Entropy, Sensors, Symmetry, Information

Hawaii International Conference on System Sciences (HICSS)

Technical Skills

Cryptographic	secure multi-party computation, secret sharing, witness encryption, threshold signature schemes, homomorphic encryption, lattice cryptography, zero-knowledge proofs, differential privacy, information theory
Languages	C/C++, Python, Rust, Bash, Lua, LaTeX
Developer	Version control (Git, SVN), CMake, Make, GDB, Valgrind, Neovim, VS Code
Platforms	Docker, AWS EC2, GitHub, HPC Clustering, Linux, Unix, Windows
Libraries	arkworks, MP-SPDZ, GMP, GMPFR, GSL, STL, OpenSSL, SageMath, NumPy, Pandas, SciPy, Matplotlib, TensorFlow

Teaching

At the **University at Buffalo**:

CSE 116 Computer science II (Instructor)	2 semesters
CSE 4/529 Algorithms for Modern Computing Systems (TA)	3 semesters
CSE 4/531 Analysis of Algorithms (TA)	1 semester
CSE 542 Software Engineering Concepts (TA)	1 semester

At **Fordham University**:

PHYS 1511/12 Physics I/II Lab (Instructor)	4 semesters